Bacterial Whole Genome Sequencing Analysis Report

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Table of Contents

Sample Extraction Protocol

Library preparation

Sequencing Protocol

Executive Data Analysis Summary

Methods – Data Analysis

Bioinformatics Pipeline

Data QC Summary

Genome Assembly

16s rRNA Extraction and Phylogenetic tree generation

Taxonomy Assignment

Bacterial Genome Annotation

Identification of AMR genes

Secondary Metabolite Annotation

Sequence Typing and goeBURST Analysis

Identification of Virulence genes

Whole Genome Circular Plot

Standard deliverables list & Access

Sample Extraction Protocol:

Bacterial pellet was added to a Qiagen PowerBead Pro Tube with 800 µl of Solution CD1 and vortexed at maximum speed for 10 minutes to lyse the cells. It was centrifuged at 15,000 × g for 1 minute and the supernatant was transferred to a new 2 ml microcentrifuge tube. 200 µl of Solution CD2 was added, vortexed for 5 seconds, centrifuged again at 15,000 × g for 1 minute, and the supernatant was transferred to a new tube. 600 µl of Solution CD3 was added, vortexed for 5 seconds, and 650 µl of the lysate was loaded onto an MB Spin Column; centrifuged and the flow-through was discarded, repeating these steps until all lysate had passed through. The column was sequentially washed with 500 µl of Solution EA and 500 µl of Solution C5, discarding the flow-through after each spin, then centrifuged at up to 16,000 × g for 2 minutes to dry the membrane. The column was placed into a clean 1.5 ml microcentrifuge tube, and DNA was eluted with 30 µl of nuclease-free water (NFW) by centrifugation at 15,000 × g for 1 minute. DNA concentration and integrity were subsequently assessed using the Qubit DNA High Sensitivity assay and agarose gel electrophoresis.

Library preparation

100 ng DNA in 26 µL was combined with 7 µL NEB Ultra II FS Reaction Buffer and 2 µL Enzyme Mix, then incubated at 37°C for 15 minutes for fragmentation and at 65°C for 30 minutes for end repair/dA-tailing. Next, 15 µL Ligation Enhancer, 30 µL Ligation Master Mix, and 2.5 µL undiluted NEBNext Adaptor were added to the 35 µL reaction, incubated at 20°C for 15 minutes, followed by the addition of 3 µL USER Enzyme for UMI adaptors and incubation at 37°C for 15 minutes. The 85.5 µL ligation mix was cleaned up with 87 µL SPRIselect/AMPure XP Beads, washed twice with 200 µL 80% ethanol, and eluted in 17 µL 0.1X TE, with 15 µL supernatant transferred. For PCR, 15 µL adaptor-ligated DNA was mixed with 5 µL Index Primer and 25 µL Q5 Master Mix, amplified with 6 cycles (98°C for 30 sec; [98°C for 10 sec, 65°C for 75 sec] × 7; 65°C for 5 min), cleaned up with 45 µL beads, and eluted in 33 µL 0.1X TE. The library was quantified, ~200–350 bp inserts were verified,

Table 1: Sample Details

Sl No.	Sample Name	Theracues Sample ID
1.	Uberis Mastitis	SAM8110

Sequencing Protocol

Prepared libraries were quantified using Qubit High Sensitivity Assay (Invitrogen, Cat# Q32851). The obtained libraries were pooled and diluted to final optimal loading concentration before cluster amplification on the Illumina Novaseq 6000 flow cell to generate 150 bp paired-end reads

Executive Data Analysis Summary

We analyzed 1 Bacteria Whole Genome Sequence Sample (SAM8110). The raw reads were filtered using Cutadapt for quality scores and adapters. Trimmed reads were further processed to check for contamination and contaminated reads were filtered out. Filtered reads were assembled using SPAdes with --isolate option. The --isolate option in SPAdes is recommended for assembling high-coverage isolate. The assembled data quality check was performed using QUAST. QUAST stands for QUality ASsessment Tool. It evaluates genome/metagenome assemblies by computing various metrics. Then we used CheckM tool to estimate genome completeness and contamination. From Taxonomy Assignment it was confirmed that the strain is Streptococcus uberis.

Bacterial Genome Annotation was identified using RASTtk webtool. We have identified Antimicrobial Resistance (AMR) Genes using ResFinder , CARD and ABRicate tool. Identification and Annotation of Secondary Metabolites were performed using antiSMASH webtool . Identification Sequence Typing was performed using pubMLST online webtool and goeBURST was performed using phyloviz tool. We have identified virulence genes using VFDB Database and generated Whole Genome Circular Plot using CG-VIEW tool.

Methods – Data Analysis

Bioinformatics Pipeline

The following bioinformatics steps were performed for analysis of the data

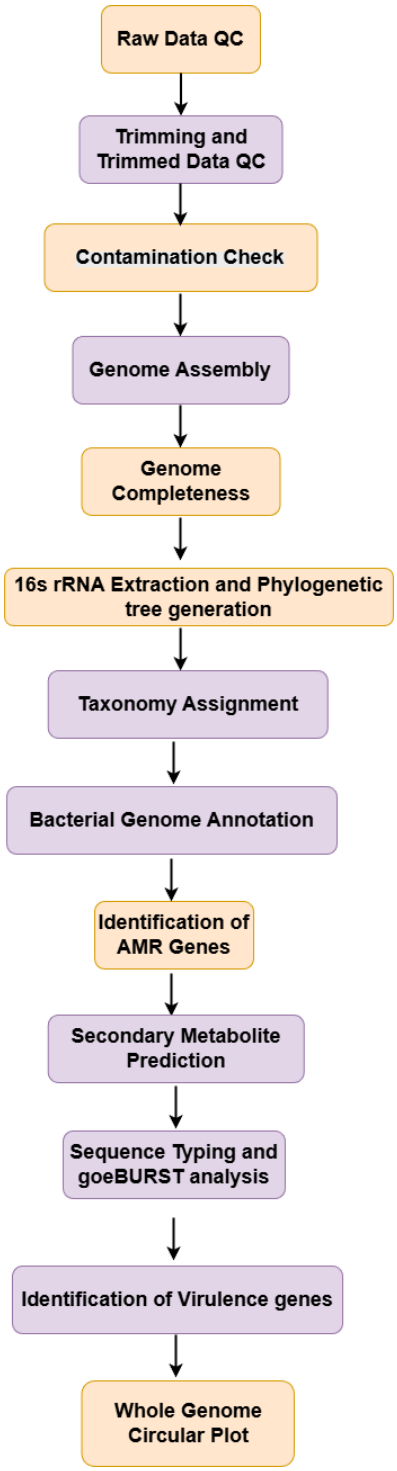

Fig 1. Bioinformatics workflow for Bacterial WGS analysis

Bioinformatics Steps

Read quality check

The following parameters from fastq file were checked as part of quality check- Base quality score distribution, Sequence quality score distribution, Average base content per read, GC distribution in the reads, PCR amplification issue, Check for over-represented sequences and Adapter trimming.

Based on the quality report of fastq files trimming of raw read was performed where necessary to only retain high quality sequence for further analysis. In addition, the low-quality sequence reads are excluded from the analysis. The adapter trimming was performed using Cutadapt (v-3.5).

The Raw reads were checked for contamination using Kraken tool. It is designed to assign taxonomic labels to short DNA sequences, enabling researchers to identify the different organisms present in a sample.

Genome Assembly

The paired-end reads are assembled using SPAdes. SPAdes is a versatile toolkit designed for assembly and analysis of sequencing data. SPAdes package contains assembly pipelines for isolated bacteria. We have used –isolate since this flag is highly recommended for high-coverage isolate. We used CheckM, an automated method for assessing the quality of a genome.

16s rRNA Extraction and Phylogenetic tree generation

16sRNA was extracted using the tool ContEst16S from EzBioCloud. By using the 16s rRNA sequence and its taxonomy hits, we have generated phylogenetic tree using MAFFT and MEGA software.

Taxonomy Assignment

The assembled reads are further processed for Taxonomic assignment to identify the species. We have used rMLST which is species identification method based on the principle of identifying the lowest common taxonomic node (LCTN) of each rMLST allele found in the bacterial genome under investigation.

We used ANI(Average Nucleotide Identity) using PubMLST and dDDH (digital DNA-DNA hybridization) which mimics traditional wet-lab DNA-DNA hybridization, which is the gold standard for prokaryotic species demarcation.

Bacterial Genome Annotation

The assembled genome scaffolds 3 samples were annotated using RASTtk webtool to ensure comprehensive functional characterization.
RAST (Rapid Annotations using Subsystems Technology) platform was employed via the RASTtk pipeline (RAST tool kit). RASTtk is a flexible and modular framework that supports accurate

structural and functional annotation of bacterial genomes based on subsystems-based classification. The tool's outputs annotated files in formats suitable for downstream analysis (GFF, GBK, and others).

Identification of AMR (Antimicrobial Resistance) genes

To investigate the antimicrobial resistance potential 3 species, AMR gene prediction was carried out using three independent and widely accepted tools: ResFinder, the Comprehensive Antibiotic Resistance Database (CARD), and ABRicate.
ResFinder, developed by the Center for Genomic Epidemiology (CGE), detects acquired resistance genes in bacterial genomes by aligning sequences to a curated AMR database.
The CARD (Comprehensive Antibiotic Resistance Database) was accessed through its Resistance Gene Identifier (RGI) tool.
ABRicate, a command-line tool which is designed to rapidly screen DNA sequences for genes associated with antimicrobial resistance.

Secondary Metabolite Annotation

To explore the secondary metabolite potential the assembled genome scaffolds were analyzed using the antiSMASH web server (version [specify if known]).
antiSMASH (antibiotics and Secondary Metabolite Analysis Shell) is a widely used genome mining tool designed for the identification, annotation, and classification of biosynthetic gene clusters (BGCs) involved in the production of secondary metabolites

Sequence Typing and goeBURST Analysis
For sequence typing, pubMLST was used. pubMLST (Public Multilocus Sequence Typing Database) supports sequence typing based on housekeeping genes specific to various bacterial species. The sequence data from the scaffold was queried against the MLST scheme, yielding a definitive Sequence Type (ST) for the isolate.

Sequence Type (ST) relationships were analyzed using the goeBURST algorithm implemented in PHYLOViZ. Allelic profiles obtained from multilocus sequence typing (MLST) were used as input, where each isolate was defined based on the combination of alleles across the seven housekeeping genes. The goeBURST method was applied to infer patterns of evolutionary descent and to identify clonal complexes based on single-locus variant (SLV) relationships.

Minimum spanning trees (MSTs) were constructed to visualize the genetic relatedness among isolates, where nodes represent individual STs and edges represent allelic differences between them. Clonal complexes were defined as groups of STs sharing high similarity, typically differing at only one allele. The analysis enabled identification of founder STs and the overall population structure of the dataset.

Identification of Virulence genes

To investigate the virulence potential , the assembled scaffold sequences were queried against the Virulence Factor Database (VFDB) using BLASTn.
The VFDB is a curated database containing known virulence genes from a wide range of bacterial pathogens. For this analysis, the nucleotide core reference dataset from VFDB was used, which includes well-characterized virulence factor sequences specific to various bacterial species

Whole Genome Circular Plot
A circular representation of sample SAM8110 Whole genome was generated using the CGView (Circular Genome Viewer) tool to visualize genomic features in an intuitive and publication-ready format.

CGView is a genome visualization platform designed to create circular plots of bacterial, plasmid, or mitochondrial genomes. It allows for detailed annotation display, including coding sequences (CDSs), RNA genes, GC content, GC skew, and other custom features. The tool accepts GenBank-formatted files and renders high-resolution visualizations that are especially useful for comparative genomics and genome organization studies.

The annotated GenBank file, prepared using RASTtk, was used as the input.

Samples and Grouping information

Following samples were sequenced and analyzed.

Table 2. Samples Information

Sl.No	Client Sample ID	TheraCUES Sample ID
1	Uberis Mastitis	SAM8110

Results

Data QC Summary

Table 3 summarizes the overall data generated and the average read quality observed acrossthe reads.

Table 3. Raw Data QC Summary

Sample Information		Before Trimming		After Trimming					
TheraCUES Sample ID	Sample Name	Total paired reads (Before adapter trimming)	Data generated (Gb)	Total paired reads (After adapter trimming)	Percentage of reads remaining after trimming (%)	Avg.base quality (phred)(phred) after trimming	Total data (reads)>= Q30% (phred) after trimming	Percentage of reads remaining after contamination removal (%)	% of reads forming Top species
SAM8110	Uberis Mastitis	600000	0.1812	588683	98.11383	39.46199	99.78392	77.91183	Streptococcus uberis (75%)

Genome Assembly Summary

The below Table 4 summarizes the assembled genome summary.

Table 4. Genome Assembly Summary

Statistics	scaffolds
# contigs	650
# contigs (>= 0 bp)	876
# contigs (>= 1000 bp)	508
# contigs (>= 5000 bp)	107
# contigs (>= 10000 bp)	9
# contigs (>= 25000 bp)	0
# contigs (>= 50000 bp)	0
Largest contig	18278
Total length	1862193
Total length (>= 0 bp)	1932846
Total length (>= 1000 bp)	1759660
Total length (>= 5000 bp)	787706
Total length (>= 10000 bp)	112524
Total length in Mb	1.9 Mb
N50	4240
N90	1334
auN	4948.6
L50	139
L90	436
GC (%)	36.49
Average Coverage	44.4x
Average Coverage to Ref Genome*	40.5x
Percent of reference genome base covered	96.5%

*Reference genome utilized throughout the entire study was S.uberis strain NCTC3858 (

GCF_900475595.1).

Table 5: Genome Completeness using CheckM tool:

Bin Id	Marker Lineage	# Genomes	# Markers	# Marker Sets	0	1	2	3	4	5+	Completeness	Contamination	Strain heterogeneity
SAM8110	Streptococcus	226	551	259	20	494	36	1	0	0	96.41	6.29	61.54

16s rRNA Extraction and Phylogenetic tree generation

16s rRNA was extracted using the tool ContEst16S from EzBioCloud. 16s rRNA based phylogenetic tree:

Fig 2: 16s rRNA Based Phylognetic Tree – SAM8110

Taxonomy Assignment

rMLST Results for Species Identification

Table 6: rMLST Results for Species Identification

Rank	Taxon	Support	Taxonomy
SPECIES	Streptococcus uberis	100%	Bacillota > Bacilli > Lactobacillales > Streptococcaceae > Streptococcus > Streptococcus uberis

Bacterial Genome Annotation

The assembled genome were successfully annotated using RASTtk pipeline to ensure comprehensive structural and functional characterization.

Identification of AMR (Antimicrobial Resistance) genes

To identify antimicrobial resistance (AMR) genes, the assembled FASTA sequence was analyzed using three independent tools: ResFinder, CARD-RGI, and ABRicate. For ResFinder and the Resistance Gene Identifier (RGI) tool from the Comprehensive Antibiotic Resistance Database (CARD), only hits with ≥95% sequence identity and ≥98% length coverage were considered as valid AMR gene predictions. Additionally, ABRicate was used to rapidly screen the genome for resistance genes by querying multiple AMR databases. The combined use of these tools ensured robust and comprehensive detection of both acquired and intrinsic resistance genes within the genome.

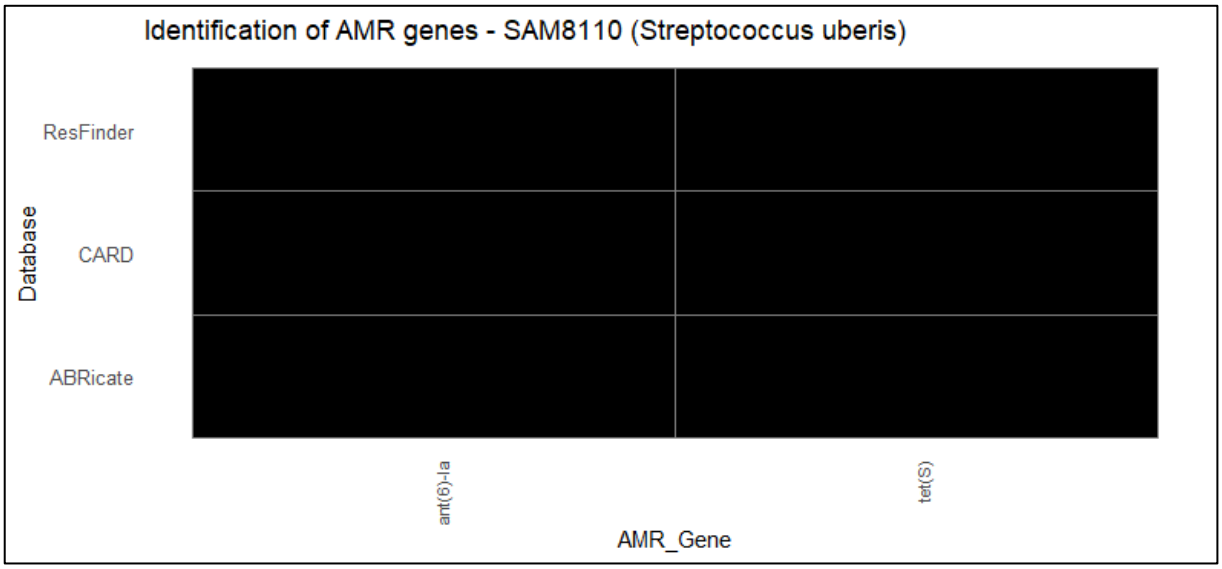

Fig 3: Identification of AMR genes by 3 tools

Sequence Typing and goeBURST Analysis

Whole genome-based sequence typing through the PubMLST website revealed the existence of novel ST (unknown according to PubMLST) sequences in the S. uberis for the sample SAM8110 and found the 5 closest profile - ST-474 , ST-984, ST-1274, ST-1731 and ST-2169.

Table 7: Sequence Type results from pubMLST tool for SAM8110

Locus	Allele	Length	Contig	Start position	End position	Locus	Allele	Length	Contig
ddl	1	357	NODE_27_length_8199_cov_8.812115	3678	4034	ddl	1	357	NODE_27_length_8199_cov_8.812115
gki	5	454	NODE_19_length_9168_cov_9.274337	3378	3831	gki	5	454	NODE_19_length_9168_cov_9.274337
tdk	17	500	NODE_15_length_9504_cov_8.640713	1570	2069	tdk	17	500	NODE_15_length_9504_cov_8.640713
tpi	1	373	NODE_213_length_3234_cov_10.513779	2209	2581	tpi	1	373	NODE_213_length_3234_cov_10.513779
yqiL	17	439	NODE_111_length_4859_cov_9.637390	1855	2293	yqiL	17	439	NODE_111_length_4859_cov_9.637390

Matching profile	
Closest profile	ST-474 , ST-984, ST-1274, ST-1731 and ST-2169
Mismatches	2
Loci matched	5/7 (71.4%)

goeBURST analysis of the Streptococcus uberis isolates was performed using allelic profiles obtained from MLST in PHYLOViZ. The minimum spanning tree revealed the population structure and evolutionary relationships among the identified Sequence Types (STs). Most STs clustered into defined clonal complexes; however, the five STs closest to our novel ST—ST-474 (India), ST-984 (Thailand), ST-1274 (China), ST-1731 (Bangladesh), and ST-2169 (country of origin unknown)—were all identified as singletons, not belonging to any clonal complex.

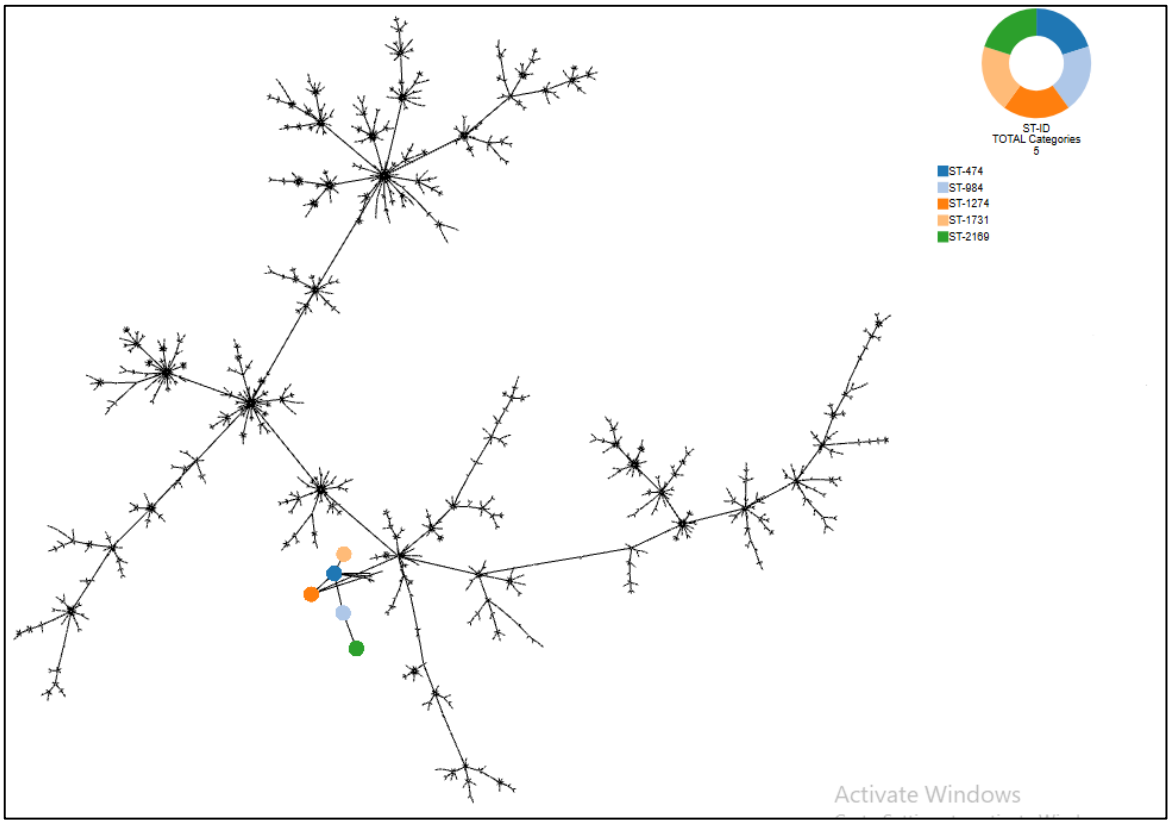

Fig 4: goeBURST analysis evolutionary tree using PHYLOViZ

Secondary Metabolite Annotation

The genome of SAM8110 was analyzed using the antiSMASH web server to investigate its potential for secondary metabolite biosynthesis. The analysis revealed the presence of 5 metabolites.

Table 8: Secondary Metabolite from antiSMASH web server.

Contig	Region	Type	From	To
NODE_7_length_10307_cov_9.114663	Region 7.1	T3PKS	1	10,307
NODE_56_length_6778_cov_8.901507	Region 56.1	Terpene precursor	1	6,778
NODE_160_length_3812_cov_11.886212	Region 160.1	RiPP-like	1	3,812

NODE_267_length_2569_cov_11.681380	Region 267.1	RiPP-like	1	2,569
NODE_409_length_1465_cov_10.045389	Region 409.1	RRE-containing	1	1,465

Identification of Virulence genes

The assembled sequences was subjected to BLAST analysis with the cutoff of identity >= 80% and Hit length >= 100bp against the Virulence Factor Database (VFDB) nucleotide core reference sequences to identify potential virulence genes.

Whole Genome Circular Plot

Whole Genome Circular Plot of 3 samples generated by CG-View

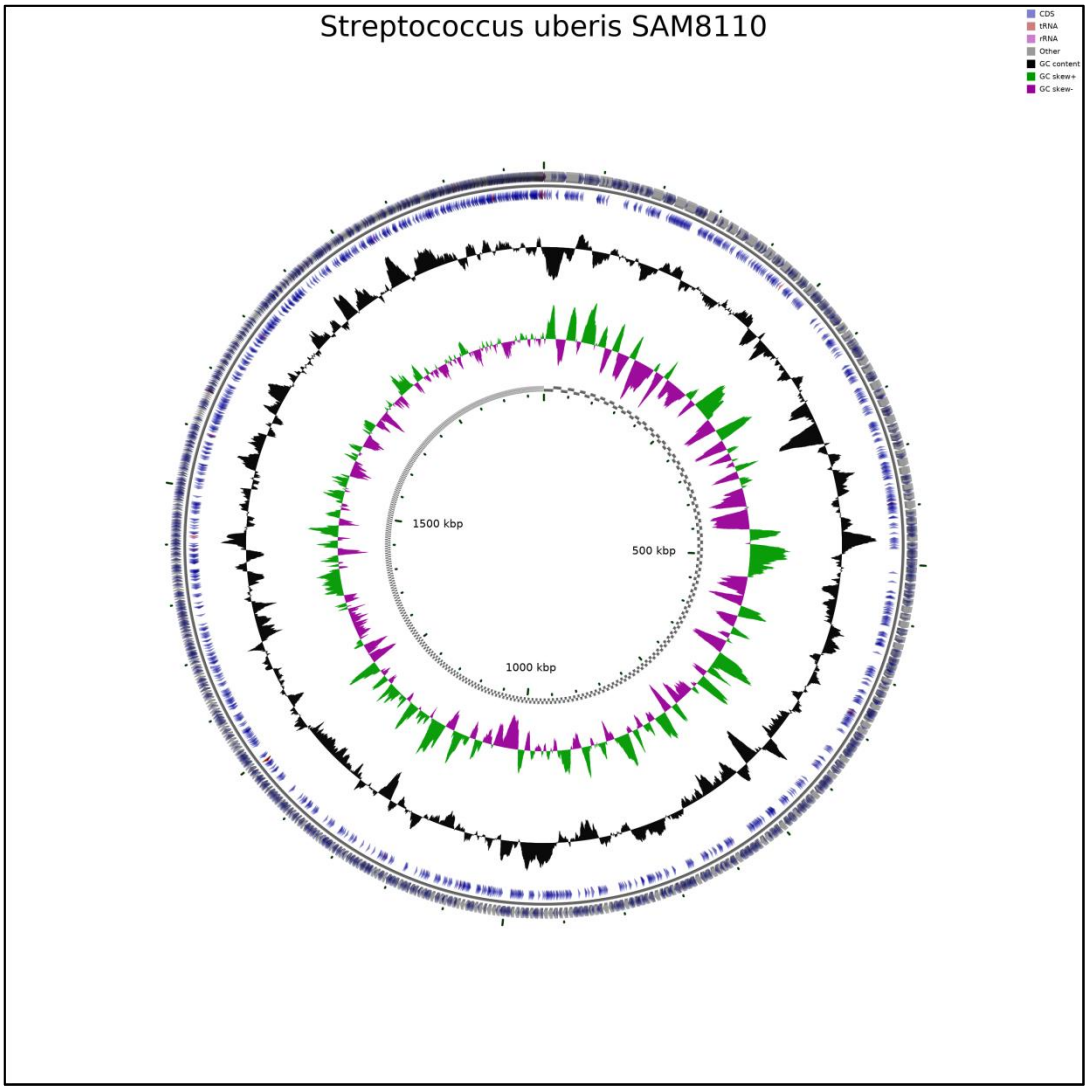

Fig 5: Whole Genome Circular Plot – SAM8110

Deliverables

Standard deliverables :

1. Raw Data and Trimmed Data Fastq files, Raw Data QC Summary Excel file and Reports for Contamination Check by Kraken
2. Genome Assembly Sequqnce in fasta format
3. Assembled Sequence QC Report by Quast and Completeness check by CheckM

4. 16s rRNA Sequence in fasta format and its taxanomic hits.
5. 16s rRNA Based Phylogenetic Tree
6. Taxanomy Assignment – Species identification by rMLST, ANI, dDDH and its resulted table
7. Bacterial Genome Annotation results by RASTtk tool
8. Antimicrobial Resistance Genes (AMR) identified by ResFinder, CARD and Abricate . Its Resulted Table and figure.
9. Sequence Type Identification Table by pubMLST webserver
10. goeBURST Analysis Result with table and evolutionary tree using PHYLOViZ tool.
11. Identification of virulence genes using VFDB database.
12. Whole Genome Circular Plot by CG-VIEW

Access to all reports :

https://workdrive.zohoexternal.com/external/566504080ebad95ff373ebdc2f190866a0120b21f33cf257e7c569710dcaab41

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